

Accelerated and Optimized Search of Imperceptible Color Vibration for Embedding Information into LCD images

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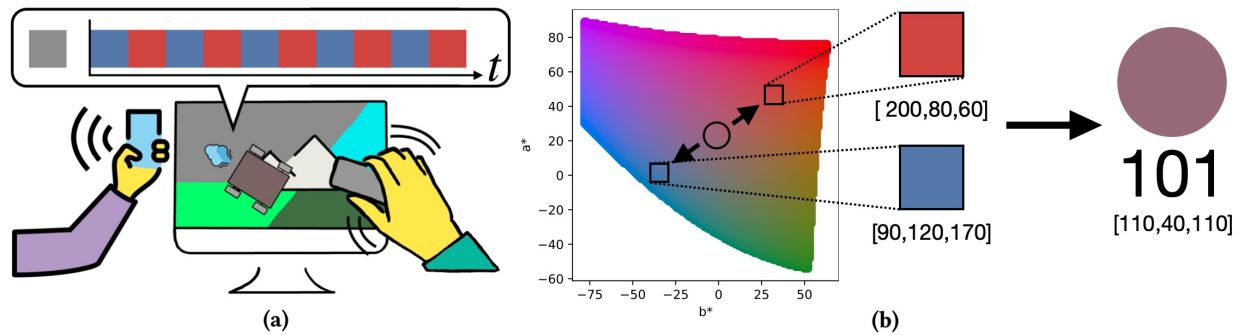


Figure 1: (a): Concept of information embedding on images using imperceptible color vibration. Devices can detect areas that appear gray to humans as switching between red and blue and acquire information on the presence of this switching to perform various actions. (b): Overview of a search method of imperceptible color vibration. Color pairs are chosen in a $L^*a^*b^*$ color space with fixed luminance (L^*), and the difference in RGB values between color pairs determines the presence of color vibration.

ACM Reference Format:

Shingo Hattori and Takefumi Hiraki. 2022. Accelerated and Optimized Search of Imperceptible Color Vibration for Embedding Information into LCD images. In *SIGGRAPH Asia 2022 Posters (SA '22 Posters)*, December 06-09, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3550082.3564190>

1 INTRODUCTION

Large, high-resolution displays are installed throughout the city as public displays. By superimposing invisible information on the images of these displays, large numbers of devices with cameras and sensors can communicate with the displays without prior pairing. Several applications have been proposed, such as operating robots or communicating information to users by displaying 2D codes on images [Sugimoto et al. 2005]. However, the display of 2D codes has the problem of compromising the appearance of displayed content.

Abe *et al.* [Abe et al. 2017] proposed a method of communicating with devices by superimposing invisible information using color vibration on images displayed on off-the-shelf liquid-crystal displays (LCD). Using this method, we can embed the information for devices in images without interfering with the displayed content (Fig. 1a).

Abe *et al.* [Abe et al. 2017] uses a simple serial loop operation to search for color pairs comprising a color vibration, which requires a very long processing time due to the huge search space.

In this paper, we propose an accelerated and optimized search method for color pairs that constitute the imperceptible color vibration for embedding information on LCD images.

To achieve fast color pair search, we parallelized the search process, which is previously done individually, by using arrays representing the amount of movement and an operation to extract elements from the array that satisfy the conditions. In addition, we investigate the amount of information that can be superimposed on nine color images using the imperceptible color vibration and clarify the applicability of embedding information into images using the color vibration.

2 METHODS

Information embedding by the imperceptible color vibration utilizes the characteristic that when the vibration frequency of two colors with fixed luminance exceeds the critical color fusion frequency (CCFF) of approximately 25 Hz, humans perceive the fused color [Jiang et al. 2007]. The color pairs used in the imperceptible color vibration are chosen to be two colors that are point-symmetric with respect to the target color in a $L^*a^*b^*$ color space with fixed luminance (Fig. 1b).

In the proposed method, we generate an array composed of a set of color pairs and extract the color pairs that satisfy the color vibration condition (cannot see by humans) and the embedding signal condition (can detect the vibration by devices).

Figure 2 shows the overview of the proposed search method.

We first decompose the target color into the L^* , a^* , and b^* channel components and create arrays that store these values.

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SA '22 Posters, December 06-09, 2022, Daegu, Republic of Korea

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ACM ISBN 978-1-4503-9462-8/22/12.

<https://doi.org/10.1145/3550082.3564190>

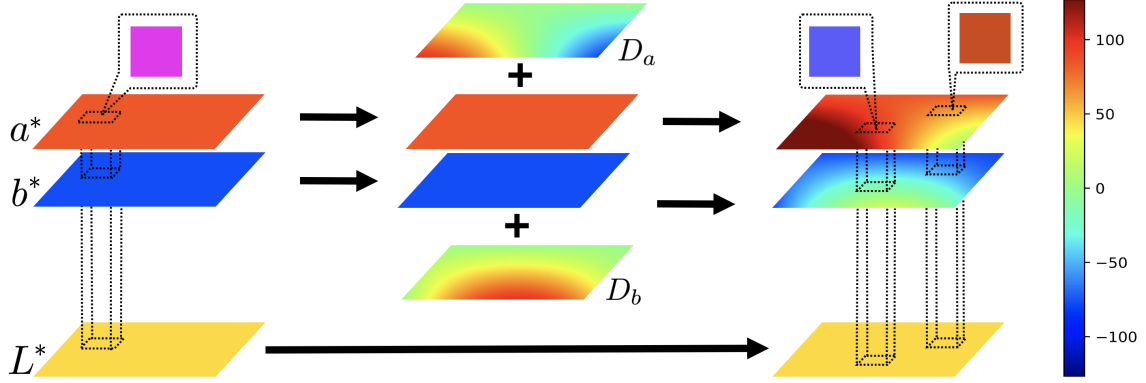


Figure 2: Overview of the proposed search method of imperceptible color vibration. We decompose the target color (purple) into a three-channel array of L^* , a^* , and b^* channels. We add the parameter arrays D_a and D_b , which represent movement in color space, to the a^* and b^* channel arrays, respectively. We can obtain multiple candidate combinations of color pairs by recombining the L^* channel array with the a^* and b^* channel arrays.

We add the arrays $D_a(D_{aij} := i \sin j)$ and $D_b(D_{bij} := i \cos j)$ to a^* array and b^* array. D_a and D_b represent the amount of movement on a^* axis and b^* axis in a $L^*a^*b^*$ color space with fixed luminance shown as Fig. 1b. Then, we recombine the added arrays with the L^* channel array to create an array of colors that have been shifted in color space from the target color. Similarly, we can create a point-symmetric color pair about the target color by subtracting D_a and D_b from a^* array and b^* array. Thanks to these processes, the proposed method enables an efficient search for color pairs in parallel by using matrix calculations.

3 EXPERIMENTS

We implemented the search for imperceptible color vibration using the previous [Abe et al. 2017] and proposed methods. We explored color vibrations using a PC (MacBook Pro 13-inch, 2019, CPU: Intel Core i5-8279U, Memory: 16 GB) for the parameter and color combinations listed in Table 1 and compared the time taken to perform them. We define a bit as high (1) if the change in the three channels (RGB) in which vibrations can be embedded in color vibration exceeds provisional perception threshold V_{th} , and low (0) if the change is below $V_{th} \times R_{novib}$.

The time required for the search was 3423.07 s for the previous method and 62.62 s for the proposed method, which means that the

Table 1: Colors, embedding information, and threshold values used in the colour pair search.

Colors (RGB)	Black (100, 100, 100), Gray (170, 170, 170), White (240, 240, 240), Red (240, 100, 100), Green (100, 240, 100), Blue (100, 100, 240), Yellow (240, 240, 100), Cyan (100, 240, 240), Magenta (240, 100, 240)
Information (RGB)	100, 010, 001, 110, 101, 011, 111
V_{th}	50, 100, 150, 200
R_{novib}	0.5, 0.25, 0.125

Table 2: The presence of color pairs that satisfy the conditions for each color and embedded information.

Information (RGB)	Black	Gray	White	Red	Green	Blue	Yellow	Cyan	Magenta
100	✓	✓	✓		✓	✓	✓	✓	✓
010									
001	✓	✓	✓	✓	✓	✓	✓	✓	✓
110		✓	✓	✓					✓
101	✓	✓	✓	✓	✓	✓	✓	✓	✓
011									
111	✓	✓	✓	✓		✓	✓		✓

proposed method was about 54 times faster. Table 2 shows the results of the search for combinations of signals that can be embedded in representative nine-color images. For the six colors with small G values (black, gray, white, red, blue, and magenta), it was found that more than four types of information can be superimposed.

4 CONCLUSION

We proposed a high-speed and optimized search method for imperceptible color vibration for embedding information in LCD images. In future work, we will evaluate whether the color vibration is perceived as the same as the original color by user experiments.

ACKNOWLEDGMENTS

This work was supported by JST ACT-X Grant Number JPMJAX1900.

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